

Masked-Media Protocol v3

A receiver-gated, signed and encrypted masked-media protocol for image, audio and video carriers.

AES-256-GCM

RSA-2048

SHA-256

1-8 LSB



ENCRYPT → SIGN → EMBED → VERIFY

Security Properties and Trust Model

01



Confidential payload

The complete record—not only the user bytes—is encrypted with AES-256-GCM.

02



Intended receiver

RSA-OAEP hides the packet location, depth, length and AES session material.

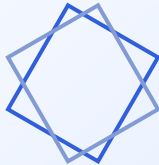
03



Trusted sender

RSA-PSS authenticates the ciphertext, carrier interpretation and embedding geometry.

04

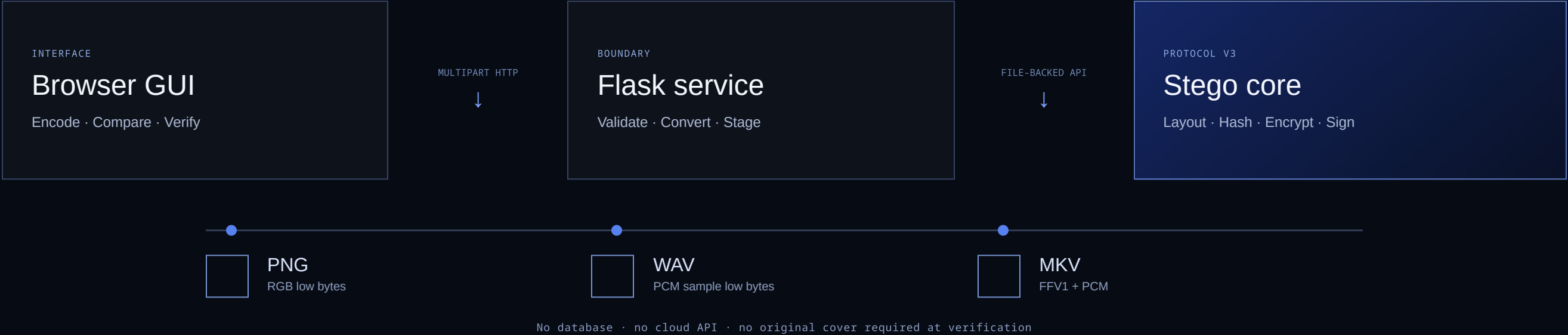


Media integrity

A reproducible masked SHA-256 hash detects changes without needing the original cover.

Payload bytes are released only when every reachable check produces **Authentic**.

Application and Protocol Architecture

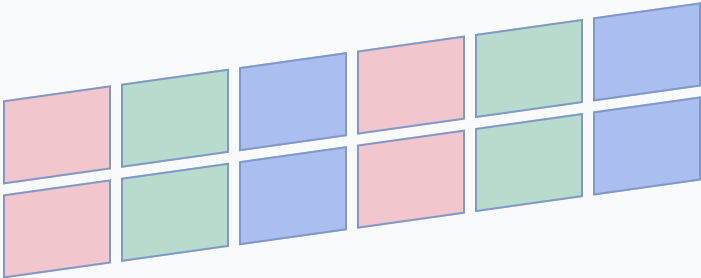


One Protocol, Three Carrier Adapters

01

PNG

IMAGE



Carrier unit: low byte of R, G or B.

Order: row-major, R → G → B.

Fixed data: alpha and 16-bit high bytes.

02

WAV

AUDIO



Carrier unit: low byte of each PCM sample.

Order: declared interleaved samples.

Fixed data: every other sample byte.

03

MKV

OPTIONAL VIDEO

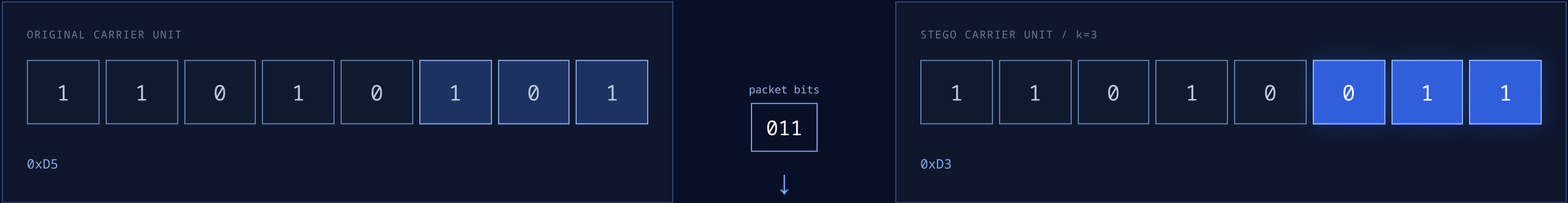


Carrier unit: RGB low bytes, then s16 audio low bytes.

Fixed data: timing, high bytes and alpha.

Output: lossless FFV1 + PCM.

Embedding Data Through LSB Replacement

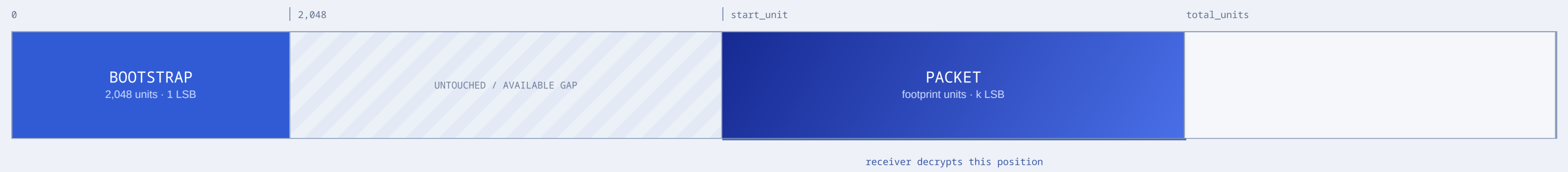


```
stego = (carrier & ~((1 << k) - 1)) | packet_field
```

Bytes become bits MSB-first; the packet field occupies the low k bits.

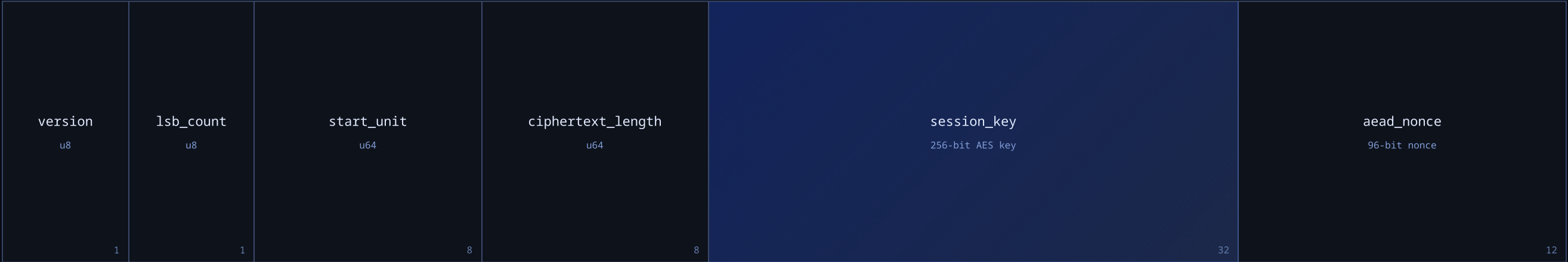


Fixed Bootstrap, Relocatable Packet



PUBLIC SHAPE Bootstrap location Always begins at unit 0. Its existence may be observable.	PRIVATE GEOMETRY Packet location + depth Recovered only by opening the RSA-OAEP bootstrap.	TAMPER BINDING Signed start unit Relocating a packet produces Signature Invalid.
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Receiver-Only Packet Discovery



AES-GCM AAD / 18 BYTES

version || lsb_count || start_unit || ciphertext_length

The geometry recovered from the bootstrap must also authenticate under the payload encryption.

Encrypted Payload Record Layout



Separate Cryptographic Security Roles



Exactly What the Sender Signs

DOMAIN INF2005-ACW1\0SIGN\0	INTERPRETATION version · media_code · k	GEOMETRY total · start · footprint · length	MEDIA CONTEXT dimensions / PCM format / video timing shape	CONTENT complete AES-GCM ciphertext + tag
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Integrity Without the Original Cover



Same logical media + different embedded bits ⇒ same masked hash.
Changed preserved media ⇒ **Tampered**.

LSB Depth Controls Capacity and Distortion

SELECT PACKET LSB DEPTH

1

2

3

4

5

6

7

8

subtle

maximum capacity

EXAMPLE PACKET

448 B

3,584 bits

FOOTPRINT

3,584

carrier units

PACKET UNIT BITS PRESERVED

87.5%

7 of 8 bits

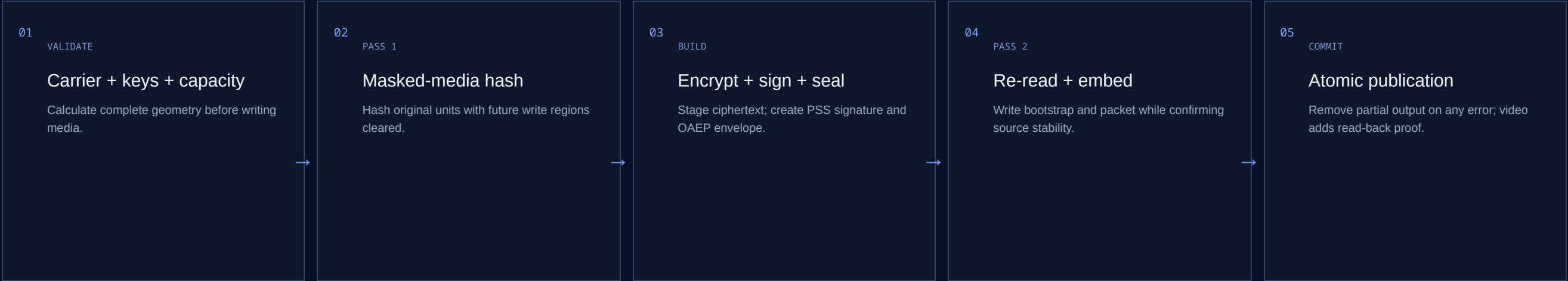
packet footprint

same packet / fewer units as k increases

$$\text{footprint} = \text{ceil}(((\text{ciphertext_length} + 256) \times 8) / k)$$

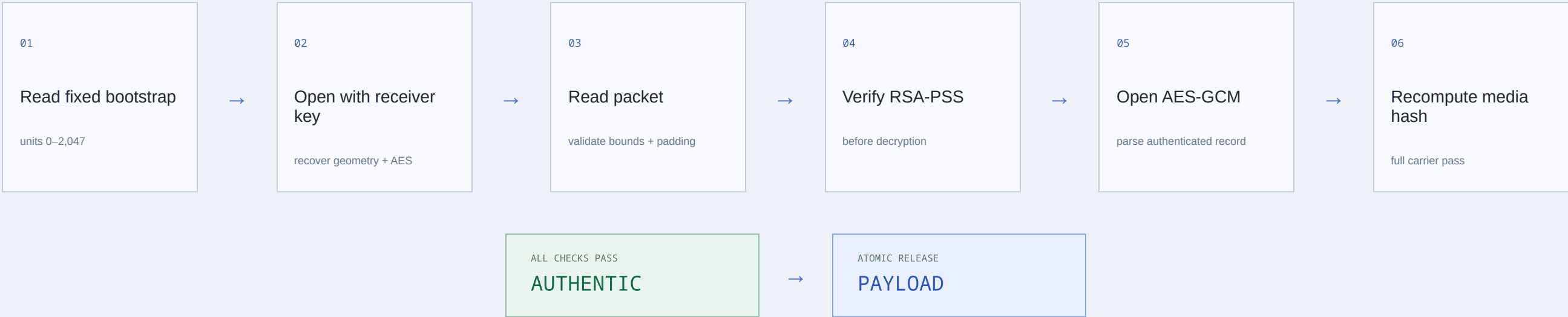
$$\text{max payload} = \text{floor}((\text{total_units} - \text{start_unit}) \times k / 8) - \text{overhead}$$

Validated Two-Pass Encoding



● Source changed between passes? **ABORT** Output differs after video mux? **ABORT**

Proof Before Plaintext Release

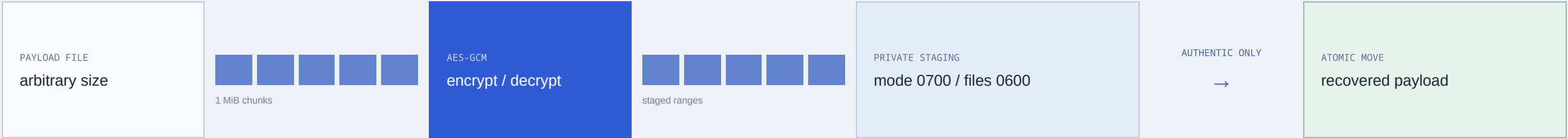


Receiver provides only: stego media · trusted sender public key · receiver private key

Defined Outcomes for Every Failure Stage

ALL GATES Authentic Signature, GCM and media hash agree.	RECEIVER_GATE Payload Missing Bootstrap cannot be opened.	GEOMETRY Wrong Start Location Packet overlaps or exceeds carrier.	SENDER_PROOF Signature Invalid PSS rejects signed bytes or layout.
	RECORD_AUTH Cannot Decrypt GCM tag rejects signed ciphertext.	MEDIA_INTEGRITY Tampered Signed record valid; media hash differs.	FORMAT_READ Cannot Verify Malformed, unsupported or incomplete.

Bounded-Memory File Processing



3 GiB reserve Upload and output operations keep shared free-space headroom.	Failure-safe No partial carrier or unauthenticated payload is published.	Crash caveat SIGKILL or power loss can leave private staging for manual cleanup.
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Canonical Carriers From Common Media Sources

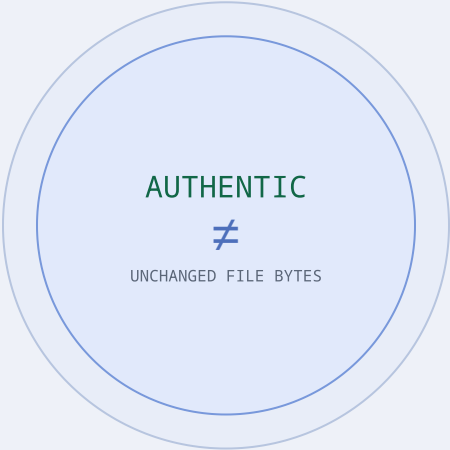


The protocol authenticates canonical decoded media—not the original compressed source bytes or encoder settings.

What an Authentic Verdict Proves

INSIDE THE BOUNDARY

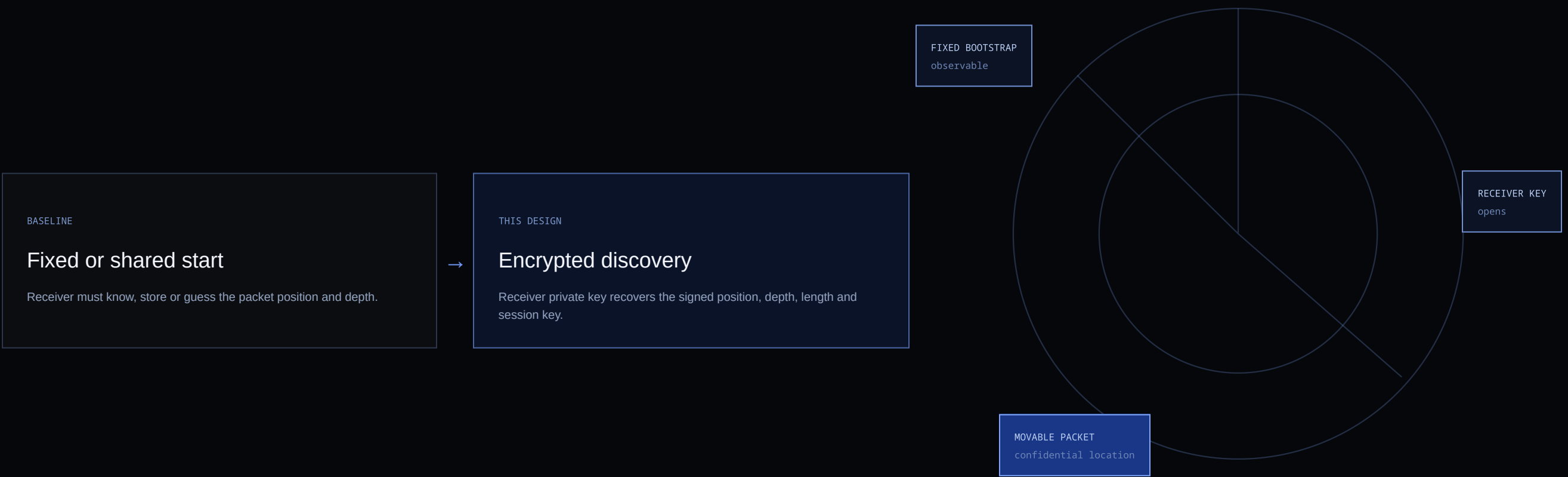
- Encrypted payload + typed metadata
- Sender-signed ciphertext and geometry
- Preserved RGB and PCM sample data
- PNG alpha and multi-byte fixed data
- Canonical video pixels, audio and timing



OUTSIDE THE BOUNDARY

- Overwritten cover LSBs
- PNG ancillary and WAV non-sample chunks
- Many container tags and display metadata
- Real-world sender identity / PKI
- Replay protection and key management

Encrypted Bootstrap Protects Packet Geometry



Evidence From Tests and Demonstration

308

Python tests passed

4

browser-boundary tests passed

1-8

LSB depths round-trip

7

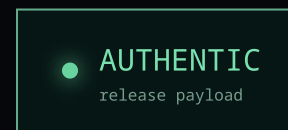
defined verification verdicts

CASE	PNG	WAV	EXPECTED
Correct keys + unchanged media	✓	✓	Authentic
Preserved media bit changed	✓	✓	Tampered
Wrong sender public key	✓	✓	Signature Invalid
Wrong receiver private key	✓	✓	Payload Missing
Payload exceeds capacity	✓	✓	Refused before write

PROTOCOL V3 / TECHNICAL DESIGN

Protocol Verified. Payload Released.

receiver access + sender authenticity + media integrity



Questions / live demonstration